

WEEK 10 PROJECT

Virtual Environment and FastAPI Setup

Create an Isolated Python Project and Export requirements.txt

Beginner-Friendly Python Assignment

Main goal: Create a separate virtual environment, install FastAPI, run one simple API, and save the installed packages in requirements.txt.

Student Name: _____	Roll Number: _____
Class / Batch: _____	Student Signature: _____

1. Project Condition

Create a simple FastAPI-ready project inside a virtual environment. The project should run without errors and should contain a requirements.txt file.

- Create a project folder named fastapi_project.
- Create a virtual environment named venv inside the project folder.
- Activate the virtual environment.
- Install FastAPI and Uvicorn.
- Create a simple FastAPI file named main.py.
- Run and test the FastAPI project in the browser.
- Export all installed packages to requirements.txt.
- Deactivate the virtual environment after finishing.

2. Simple Project Rules

Rule	Condition
Project folder	All project files should be kept inside fastapi_project.
Virtual environment	The folder should be named venv.
Activation	Install packages only after the environment is active.
Python file	The FastAPI application should be saved as main.py.
Requirements file	Create requirements.txt after installing packages.
Testing	The home route and /docs page should open successfully.

3. Final Project Structure

```
fastapi_project
|
|-- venv
|-- main.py
|-- requirements.txt
```

4. Virtual Environment in Easy Words

A virtual environment is a separate space for one Python project. Packages installed in this space do not disturb other Python projects on the computer.

Easy example: Each project gets its own box of Python packages.

5. Concepts and Commands Required

Concept / Command	Purpose
<code>python -m venv venv</code>	Creates a virtual environment named venv.
<code>venv\Scripts\activate</code>	Activates the environment on Windows Command Prompt.
<code>pip install</code>	Installs packages inside the active environment.
<code>FastAPI()</code>	Creates the FastAPI application.
<code>@app.get()</code>	Creates a simple GET route.
<code>uvicorn</code>	Runs the FastAPI application.
<code>pip freeze</code>	Shows installed packages and their versions.
<code>requirements.txt</code>	Stores the package list for the project.
<code>deactivate</code>	Closes the active virtual environment.

6. Instructions for Students

1. Create a folder named `fastapi_project`.
2. Open the folder in VS Code.
3. Open Terminal > New Terminal.
4. Create the virtual environment.
5. Activate the virtual environment.
6. Confirm that `(venv)` appears in the terminal.
7. Install FastAPI and Uvicorn.
8. Create `main.py` and write the simple API code.
9. Run the project using Uvicorn.
10. Test the API in the browser and open `/docs`.
11. Stop the server and create `requirements.txt`.
12. Deactivate the environment.

7. Create the Project Folder

```
mkdir fastapi_project
cd fastapi_project
code .
```

After code `.` runs, the project folder will open in VS Code.

8. Create and Activate the Virtual Environment

Step 1: Create the Environment

```
python -m venv venv
```

A new folder named venv will appear inside the project folder.

Step 2: Activate on Windows Command Prompt

```
venv\Scripts\activate
```

After activation, the terminal should begin with:

```
(venv)
```

Optional: Activate on macOS or Linux

```
source venv/bin/activate
```

Important: Do not install FastAPI until (venv) is visible in the terminal.

9. Install FastAPI and Uvicorn

```
python -m pip install fastapi uvicorn
```

FastAPI creates the API. Uvicorn runs the API on the computer.

10. Check Installed Packages

```
python -m pip list
```

The list should include fastapi and uvicorn.

11. Create main.py

Create a file named main.py and write the following complete code:

```
from fastapi import FastAPI

# Create the FastAPI application
app = FastAPI()

# Home route
@app.get("/")
def home():
    return {
        "message": "My FastAPI project is working"
    }

# Student route
@app.get("/student")
def student():
    return {
        "name": "Ali",
        "course": "Python"
    }
```

12. Easy Explanation of the Code

Code Part	Easy Meaning
<code>from fastapi import FastAPI</code>	Imports FastAPI into the Python file.
<code>app = FastAPI()</code>	Creates the API application.
<code>@app.get("/")</code>	Creates the home route.
<code>def home():</code>	Runs when the home route is opened.
<code>return {...}</code>	Sends JSON data to the browser.
<code>@app.get("/student")</code>	Creates another route for student information.

13. Run the FastAPI Project

```
uvicorn main:app --reload
```

Part	Meaning
main	The Python filename main.py.
app	The FastAPI object inside main.py.
--reload	Restarts the server when code is changed.

14. Test in the Browser

Open these addresses one by one:

```
http://127.0.0.1:8000
http://127.0.0.1:8000/student
http://127.0.0.1:8000/docs
```

Expected Home Result

```
{
  "message": "My FastAPI project is working"
}
```

Expected Student Result

```
{
  "name": "Ali",
  "course": "Python"
}
```

The /docs page is the automatic FastAPI testing page.

To stop the server, press:

```
ctrl + c
```

15. Create requirements.txt

After stopping the server, run:

```
python -m pip freeze > requirements.txt
```

This command creates requirements.txt and saves the installed package names and versions.

Example requirements.txt

```
fastapi==version-number  
pydantic==version-number  
starlette==version-number  
uvicorn==version-number
```

The version numbers may be different on each computer. This is normal.

16. Install the Project on Another Computer

A different user can install the same packages by running:

```
python -m venv venv  
venv\Scripts\activate  
python -m pip install -r requirements.txt
```

17. Deactivate the Environment

```
deactivate
```

The (venv) text will disappear from the terminal.

18. Complete Windows Command Sequence

```
mkdir fastapi_project  
cd fastapi_project  
code .  
python -m venv venv  
venv\Scripts\activate  
python -m pip install fastapi uvicorn  
uvicorn main:app --reload  
python -m pip freeze > requirements.txt  
deactivate
```

19. Common Problems and Easy Solutions

Problem	Easy Solution
python is not recognized	Check that Python is installed and reopen VS Code.
(venv) does not appear	Run venv\Scripts\activate again in Command Prompt.
No module named fastapi	Activate venv and install fastapi again.
uvicorn is not recognized	Activate venv and install uvicorn again.
The page does not open	Check that the server is still running in the terminal.
Port is already in use	Stop the old server using Ctrl + C, then run again.
requirements.txt is empty	Activate venv, install packages, and run pip freeze again.

20. Important Tests

Test	Expected Result
Activate venv	(venv) appears in the terminal.
Run the server	Uvicorn shows http://127.0.0.1:8000.
Open /	The working message appears.
Open /student	Student JSON data appears.
Open /docs	FastAPI documentation page opens.
Create requirements.txt	The file contains package names and versions.
Deactivate venv	(venv) disappears from the terminal.

21. Final Assignment Checklist

Check	Requirement
☐	The project folder is named fastapi_project.
☐	The virtual environment folder is named venv.
☐	The environment activates successfully.
☐	FastAPI and Uvicorn are installed inside venv.
☐	main.py contains two simple GET routes.
☐	The API runs without errors.
☐	The /docs page opens successfully.
☐	requirements.txt has been created.
☐	The student has tested the complete project.

Submission: main.py and requirements.txt. Do not submit the venv folder unless the trainer asks for it.